CHAPTER 3

Switching & Performance Metrics

Circuit · packet · virtual circuit · the four delays · RTT · throughput · bandwidth-delay product

QUICK TRICK

Trans delay $\mathbf{T_t = L / R}$ (bits ÷ rate). Prop delay $\mathbf{T_p = d / v}$ (distance ÷ speed). RTT ≈ $2T_p$. Pipe size (BDP) = **Bandwidth × RTT**. Packets over N links: $\mathbf{(N + P - 1) \times L/R}$.

3.1 Three ways to move data

● **LEVEL 1 · BEGINNER**

Circuit switching: set up a dedicated path first (like a phone call). Resources are reserved for the whole call, so no queuing, but idle time is wasted. **Packet (datagram) switching:** cut data into packets, each routed on its own with store-and-forward. **Virtual circuit:** set up a path first, then every packet carries a short VC number and follows the same path.

Property	Circuit	Datagram	Virtual circuit
Setup phase	Yes	No	Yes
Resources	Reserved	On demand	On demand (or reserved)
Address in packet	none (slot/frequency)	Full destination address	Short VC number
Path	Fixed	Each packet may differ	Fixed for the connection
Order	In order	May be out of order	In order
Router state	Per circuit	None per flow	Per VC
Congestion	No (blocked at setup)	Possible, queuing	Possible
Delay	Setup + transmit	Variable	Setup + low per-packet

QUICK TRICK

Packet switching shares links by **statistical multiplexing**. Example (Kurose): a 1 Mbps link and users who need 100 kbps but are active only 10% of the time. Circuit switching supports 10 users. Packet switching supports 35 users with a chance of ≈ 0.0004 that more than 10 are active together.

3.2 The four delays

● **LEVEL 2 · EXAM**

Delay	Formula	Depends on
Transmission T_t	L / R	Packet size and link rate
Propagation T_p	d / v	Distance and signal speed (≈ 2×10^8 m/s in cable; 3×10^8 in air)
Processing	given	Router work (header check, lookup)
Queuing	given	Traffic load

Total nodal delay = processing + queuing + transmission + propagation. **Example.** 1000 B over 100 Mbps and 3000 km at 2×10^8 m/s: $T_t = 8000/10^8 = 80$ µs; $T_p = 3\times10^6/2\times10^8 = 15$ ms. The packet is fully received after 15 ms + 80 µs = **15.08 ms**.

3.3 Store-and-forward over several links

● **LEVEL 2 · EXAM**

A router must receive the **whole packet** before sending it on. For N links of equal rate R (N − 1 routers):

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One packet of L bits      :  time = N * (L/R)          (+ sum of propagation delays)
P packets, each L bits    :  time = (N + P - 1) * (L/R) (pipelining, + propagation)
Circuit switching message :  time = setup + M/R + propagation
```

Splitting a message into packets. Message M bits, k hops, header h bits, payload p bits per packet: time $\propto (p + h)(M/p + k - 1)/R$. This is smallest at **$p = \sqrt{(M \cdot h / (k - 1))}$**.

3.4 RTT, throughput, bandwidth-delay product

● LEVEL 2 · EXAM

- **RTT** = time for a bit to go to the receiver and an acknowledgement to return = $2T_p$ (plus transmission/queuing if stated).
- **Throughput** = bits actually delivered per second. Over a path it is limited by the **slowest link** (bottleneck).
- **Bandwidth-delay product (BDP)** = bandwidth × RTT = bits "in the pipe". To keep the link 100% busy the sender needs a window of at least BDP.
- **Utilisation** = (data sent per RTT) ÷ (BDP).

COMMON MISTAKE

- Mixing units: km vs m, bits vs bytes, ms vs s.
- Using 2^{20} for Mbps. **Rates use powers of 10**: 1 Mbps = 10^6 bps. File sizes (KB, MB) follow the question text.
- Using one-way delay when the question needs RTT (and the reverse).
- Adding T_p for every packet when pipelining. Propagation is added **once** per link.

GATE QUICK CHECK — Chapter 3

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Tt = L/R           Tp = d/v           RTT = 2*Tp
BDP = R * RTT      Throughput = min(link rates) for a path
N links, P packets : (N + P - 1) * L/R        optimal packet p = sqrt(M*h/(k-1))
Setup phase        : circuit and virtual circuit only
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Transmission delay of a 1500 B packet on a 10 Mbps link.

Q2. Propagation delay for 2000 km at 2×10^8 m/s.

Q3. Which switching methods need a setup phase?

Q4. Which of datagram and virtual circuit guarantees in-order delivery?

● LEVEL 2 · EXAM

Q5. BDP of a 1 Gbps link with RTT 20 ms.

Q6. A 4000-bit message crosses 3 links of 1 Mbps each (no propagation, no header). Time as one packet, then as 4 packets of 1000 bits.

Q7. A path has links of 10, 2 and 5 Mbps. Throughput, and time to send an 8 Mb file?

Q8. 1000 B over 100 Mbps and 3000 km ($v = 2\times10^8$ m/s): time until fully received?

● LEVEL 3 · GATE

Q9. Message M = 20000 bits, header h = 50 bits, k = 5 hops. Optimal payload size per packet?

Q10. A 1 Mbps link is shared by circuit switching so that each user gets 250 kbps. How many users can be admitted?

Answers

Q	Solution
1	$12000 / 10^7$ = **1.2 ms**.
2	$2\times10^6/2\times10^8$ = **10 ms**.
3	**Circuit** and **virtual circuit**. Datagram does not.
4	**Virtual circuit** (fixed path).
5	$10^9 \times 0.02 = 2\times10^7$ bits = **2.5 MB** (2.5×10^6 B).
6	One packet: 3 × 4 ms = **12 ms**. Four packets: (3 + 4 − 1) × 1 ms = **6 ms**.
7	Bottleneck = **2 Mbps**; time = 8/2 = **4 s**.
8	80 µs + 15 ms = **15.08 ms**.
9	p = √(50 × 20000 / 4) = √250000 = **500 bits**.
10	1000/250 = **4 users**.

Next: Chapter 4 — Error Detection (parity, checksum, CRC).

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